



Deep learning methods for proteome-scale interaction prediction

Min Su Yoon^{1,a}, Byunghyun Bae^{2,3,a}, Kunhee Kim^{1,a},
 Hahnbeom Park^{3,4} and Minkyung Baek¹

Proteome-scale interaction prediction is essential for understanding protein functions and disease mechanisms. Traditional experimental methods are often limited by scale and complexity, driving the need for computational approaches. Deep learning has emerged as a powerful tool, enabling high-throughput, accurate predictions of protein interactions. This review highlights recent advances in deep learning methods for protein–protein and protein–ligand interaction screening, along with datasets used for model training. Despite the progress with deep learning, challenges such as data quality and validation biases remain. We also discuss the increasing importance of integrating structural information to enhance prediction accuracy and how structure-based deep learning approaches can help overcome current limitations, ultimately advancing biological research and drug discovery.

Addresses

¹ Department of Biological Sciences, Seoul National University, Seoul 08826, Republic of Korea

² Department of Chemistry, Seoul National University, Seoul 08826, Republic of Korea

³ Biomedical Research Division, Korea Institute of Science and Technology, Seoul 02792, Republic of Korea

⁴ KIST-SKKU Brain Research Center, SKKU Institute for Convergence, Sungkyunkwan University, Suwon 16419, Republic of Korea

Corresponding authors: Baek, Minkyung (minkbaek@snu.ac.kr);
 Park, Hahnbeom (hahnbeom@kist.re.kr)

✉ (Park H.), ✉ (Baek M.)

^a These authors contributed equally to this work.

understanding the molecular basis of disease, identifying potential therapeutic targets, and developing new drugs. Traditional experimental methods for studying protein interactions, however, are often labor-intensive, time-consuming, and limited in scale. To address these limitations, template-based computational methods have been developed [1,2]. However, their accuracy is often constrained, and they rely on expanding known interactions, limiting their applicability to interactions without experimental precedents. This creates a significant need for high-throughput, proteomic-scale interaction screening methods that can efficiently map the vast landscape of protein interactions within a cell.

Deep learning has emerged as a transformative tool in this context, offering the potential to overcome many of the limitations associated with conventional experimental screening methods. Deep learning-driven approaches can analyze large datasets with unprecedented speed and accuracy [3,4], identify novel interaction patterns, and predict interactions that may not be easily detectable through experimental techniques alone [5–7]. By integrating deep learning into proteomic-scale screening, researchers can accelerate the discovery of new protein functions, gain deeper insights into the molecular underpinnings of diseases, and streamline the drug development process.

In this review, we focus on deep learning models that can be applied for proteome-scale screening in two key areas: protein–protein interactions (PPI) and protein–ligand interactions (PLI). Major methods are summarized in Table 1. The main aim of PPI prediction is to identify and understand how proteins interact with each other within a biological system. These interactions are crucial for most cellular processes, such as signal transduction, metabolic pathways, cell communication, and immune responses. By mapping these interactions, researchers can construct detailed biological networks, infer the functions of uncharacterized proteins, and uncover mechanisms underlying various diseases. Given the complexity and expense of experimental methods, computational approaches to PPI prediction are invaluable for accelerating experimental validation and deepening our understanding of biological processes.

Current Opinion in Structural Biology 2025, **90**:102981

This review comes from a themed issue on **Artificial Intelligence (AI) Methodologies in Structural Biology (2025)**

Edited by **Chaok Seok** and **Pratyush Tiwary**

For complete overview of the section, please refer the article collection - [Artificial Intelligence \(AI\) Methodologies in Structural Biology \(2025\)](#)

Available online 22 January 2025

<https://doi.org/10.1016/j.sbi.2024.102981>

0959-440X/© 2024 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Introduction

Proteins function through complex networks of interactions that regulate virtually all biological processes. Deciphering these interaction networks is essential for

Table 1

Summary of deep learning methods for PPI and PLI screening since 2021.

For protein–protein interactions (PPIs)		
Methods	Prediction of complex structures alongside PPI prediction?	Relevant references
D-SCRIPT	No	[8]
RAPPPID	No	[9]
Topsy-Turvy	No	[10]
TT3D	No	[11]
TUnA	No	[12]
Pre-PPI	Yes	[13]
ZEPPi	Yes	[14]
AlphaFold series	Yes	[3,15–18]
RoseTTAFold series	Yes	[4,5,7]
For protein–ligand interactions (PLIs)		
Methods ^a	Explicit utilization of receptor structures?	Relevant references
ConPlex	No	[19]
DrugCLIP	No	[20]
Kinase-CPI-reanalyzes	No	[21]
HyperAttentionDTI	No	[22]
DrugBan	No	[23]
DeepPS	Yes	[24]
STAMP-DPI	Yes	[25]
AttentionSiteDTI	Yes	[26]

^a Methods not applied to screening study (e.g. AlphaFold3) were not included.

Interaction with target small-molecules is another interest in proteome-level screening. The main aim in PLI is to identify receptors that interact with a given small-molecule ligand. Receptor screening can be useful in drug repurposing, allowing researchers to take advantage of existing ADME (absorption, distribution, metabolism, and excretion) profiles of a drug that have already been experimentally analyzed. Moreover, receptor screening facilitates the early identification of potential toxicities of the small molecule, which are a leading cause of clinical trial failures. The significance of receptor screening has not been emphasized much in contrast to the ligand virtual screening.

Bioassay and database

Databases related to PPI or PLI are shown in Table 2. In the table, the names of the datasets, the number of data entries, applicability to protein–protein (PP) or protein–ligand (PL), the latest version, provided data format, and their URLs are listed. These datasets can be employed for validating computational approaches or for training networks. While PDB, BioGRID, and STRING have been the most standard databases for studying general PPIs, such datasets do not exist in the PLI domain, requiring their own dataset construction or borrowing from previous work.

The main source of datasets comes from a variety of bioassays. High-throughput experimental techniques such as yeast two-hybrid (Y2H), affinity-purification

mass spectrometry (APMS) and coimmunoprecipitation (co-IP) have provided large datasets of PPIs in several organisms [31,43,44]. In addition to these experimental methods, PPI databases expand the known interaction space by inferring potential interactions using co-expression, colocalization, homology, and text mining. Proteome-level PLI data are collected from proteome-level assays, such as affinity-based protein profiling (ABPP) [45] and drug affinity responsive target stability (DARTS) [46]. Target-specific assays provide interaction data between a particular protein family and potential ligands, such as the multiplexed inhibitor beads (MIB) assay and kinase activity assay for kinome profiling (KAYAK) for kinases [47], as well as the radiometric binding assay and the homogeneous time-resolved fluorescence (HTRF) IP1 assay for GPCRs [48]. Also, for both PPI and PLI, individual assay data from various groups comprise large portions in generic datasets. Of these, subsets passing their own criteria are collected and deposited in the datasets listed in the table.

Proteome-wide screening for protein–protein interactions

Figure 1 illustrates two common approaches for PPI prediction using deep learning-based methods. One major approach involves building a deep learning-based classifier that combines various features from pretrained protein language models, precomputed structural databases, or known experimental data. The other major approach utilizes protein structure prediction methods

Table 2

Available database type and quantity for protein interactions.

General databases						
Database	Number of entries	Entries (protein)	Applicability	Version (date)	Data type	URL
PDB [27]	224 k	220 k	PP, PL	2024.09	Structure	https://www.rcsb.org/
UniProt [28]	246 M	246 M	PP, PL	2024.04	Sequence	https://www.uniprot.org/
BioGRID [29]	2.8 M (PP) 31 k (PL)	89.8 k (PP) 2.7 k (PL)	PP, PL	4.4.237 (2024.9)	Sequence	https://thebiogrid.org/
STRING [30]	20 B	59 M	PP	12.0 ver (2023.7)	Sequence	https://string-db.org/
HuRI [31]	64 k	9 k	PP	2020	Sequence	http://www.interactome-atlas.org/
ChEMBL [32]	1.8 M ^a	15 k	PL	ChEMBL34 (2024)	SMILES	https://www.ebi.ac.uk/chembl/
PubChem [33]	119 M ^a	248 k	PL	2024.09	Structure	https://pubchem.ncbi.nlm.nih.gov/
DrugBank [34]	17 k ^a	22 k	PL	2024.09	SMILES	https://go.drugbank.com/
Secondary databases ^c						
PDBbind [35]	157 k (PDB) (PL: 19.4 k, PP: 2.8 k)	3.4 k	PP, PL	v.2020	Structure	http://www.pdbbind.org.cn/
Q-BioLiP [36]	2.9 M (PL: 488 k, PP: 32.7 k)	212 k	PP, PL	2024-08-30 ver	Structure	https://yanglab.qd.sdu.edu.cn/Q-BioLiP/
Davis [37]	72 k ^a	442	PL	2011	Sequence & SMILES	https://doi.org/10.1038/nbt.1990
STITCH [38]	1.6 B	9.6 M	PL	5.0 ver (2016)	Sequence	http://stitch.embl.de/
GalaxyDB [25]	381 k ^b	372	PL	2022	Structure	https://github.com/biomed-AI/STAMP-DPI?tab=readme-ov-file
PLINDER [39]	449 k ^{b,c}	74k ^d	PL	2024.06	Structure	https://www.plinder.sh
BindingDB [40]	2.9 M 1.3 M ^a	9.3 k	PL	2024.09	Sequence & SMILES	https://www.bindingdb.org/rwd/bind/index.jsp
LIT-PCBA [41]	7.7 k ^b	15	PL	2024.09	Structure	https://drugdesign.unistra.fr/LIT-PCBA/
DUD-E [42]	22 k ^b	102	PL	2024.09	Structure	https://dude.docking.org/

^a Number of compounds.^b Number of active compounds.^c Collected by processing general primary databases.^d Number of redundant set.

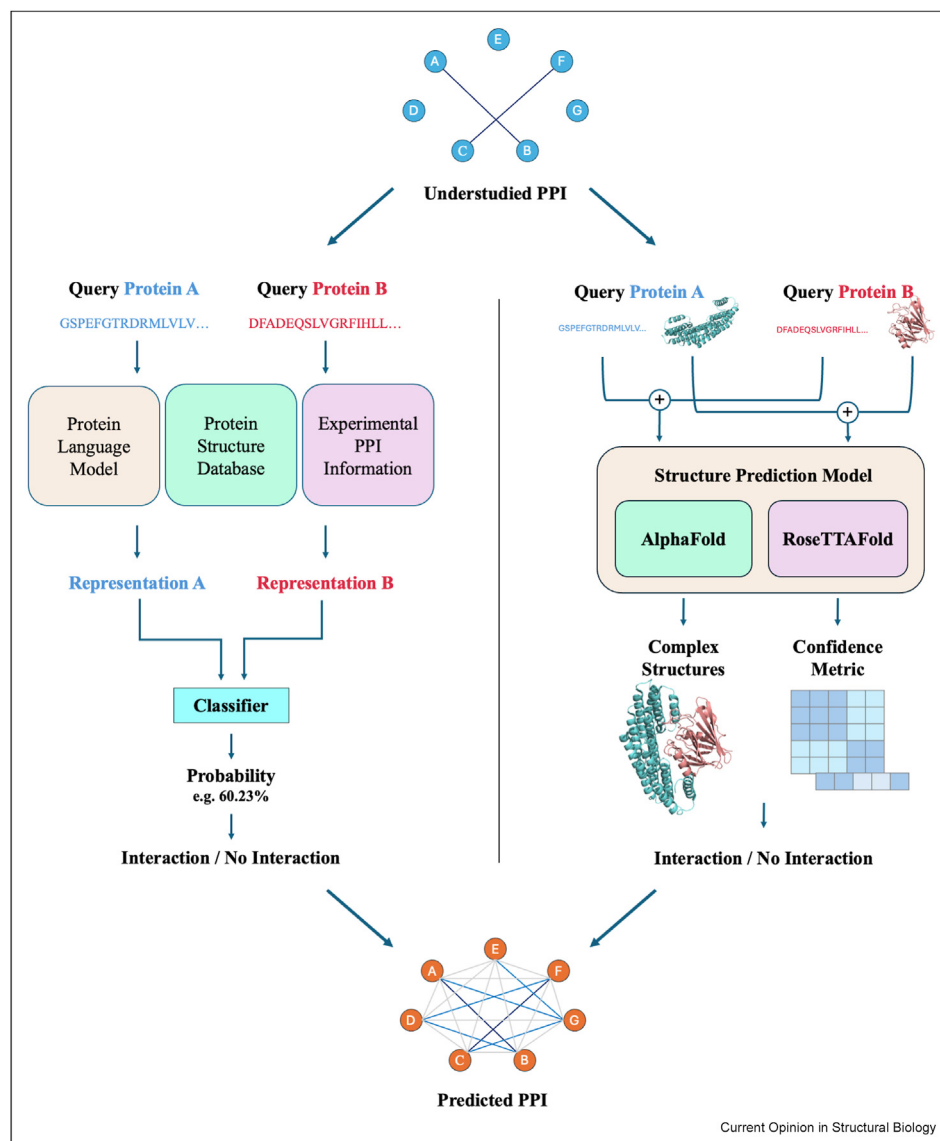
such as AlphaFold and RoseTTAFold. Structure prediction methods roughly require 1000 times more computational resources than classifier-based methods. Minor modifications [5,7], however, have enabled these methods to be applied on a genomic scale to effectively screen for PPIs.

While PIPR [49] and RAPPPID [9] directly predict the interaction probability of two given proteins based on protein sequence, D-SCRIPT [8] and TUnA [12] leverage pretrained protein language models, such as the Bepler and Berger model [50] and Evolutionary Scale Model-2 (ESM-2) [51], to enhance feature extraction from amino acid sequences. Topsy-Turvy [10] builds upon D-SCRIPT by complementing sequence-based approaches with network-based methods, incorporating known PPI networks from well-studied species by calculating GLIDE [52] scores to improve prediction

accuracy. KSGPPI [53] takes hybrid approaches that incorporate ESM-2 pretrained protein language model as well as known interaction network information retrieved from STRING database.

Since high-accuracy protein structure prediction methods such as AlphaFold [3] and RoseTTAFold [4] became available, many deep learning models have been developed that utilize structural information to predict PPIs. Topsy-Turvy 3D (TT3D) [11] further extends Topsy-Turvy to take structural features by jointly modeling both amino acid sequence and the 3D interaction alphabet (3Di) sequence [54], which encodes structural information of distances and angles along the protein backbone. PrePPI [13] takes inspiration from template-based modeling, predicting PPIs using features extracted from template complex structures that share structural similarities to query proteins. ZEPPI

Figure 1



Schematic view of current protein–protein interaction prediction methods.

[14], an extension of PrePPI, utilizes paired multiple sequence alignments to verify whether the proposed interface aligns with the coevolutionary information.

Other approaches have focused on extending protein structure prediction methods to predict PPIs. Coevolution prediction methods, such as EVcomplex2 [55], have laid the groundwork for identifying evolutionary residue couplings that indicate potential interactions. Building on this, RoseTTAFold and AlphaFold, which can capture coevolutionary patterns, have been utilized to predict PPIs. Since RoseTTAFold and AlphaFold were first utilized to predict yeast interactomes [5],

many methods have applied these models and their variants, such as RF2-Lite [7], AlphaFold-multimer [15], and FoldDock [16], to map interaction networks in various systems, including human mitochondrial proteins [6], human pathogen proteins [7], African swine fever virus proteins [56], cancer driver proteins [57], and Drosophila proteins [17]. In these studies, various metrics, such as interchain contact probability [5], mean interchain predicted aligned error (pAE) [58], predicted DockQ score (pDockQ) [16], and machine learning classifiers [18], are applied to assess the likelihood of an interaction. This protein structure prediction AI-based interaction screening method offers the

distinct advantage of predicting not only whether proteins interact but also the specific structures of their binding interfaces.

Proteome-wide screening for ligand-binding receptors

In this section, we examine the current state of ligand-binding receptor screening methods. Screening methods are primarily classified along the receptor structural information utilization, with an additional focus on core approach (traditional vs DL-based) (Figure 2).

Ligand similarity-based approaches

A group of methods categorized as ligand-based methods use the ligand information as the major input. They are based on the ligand similarity search on known complexes with an assumption that similar ligands are highly likely to act on similar proteins. The traditional research direction was to choose ligand information types (i.e. molecular fingerprints) optimal for the similarity comparison. These methods took raw similarity metrics or simple machine learning models to combine them to predict ligand-binding receptors.

Since the emergence of DL-based approaches, it became common to take protein sequences as an additional input to the ligand information. These types of studies are often called drug–target interaction (DTI) [19–21,59,60]. A group of methods were published with

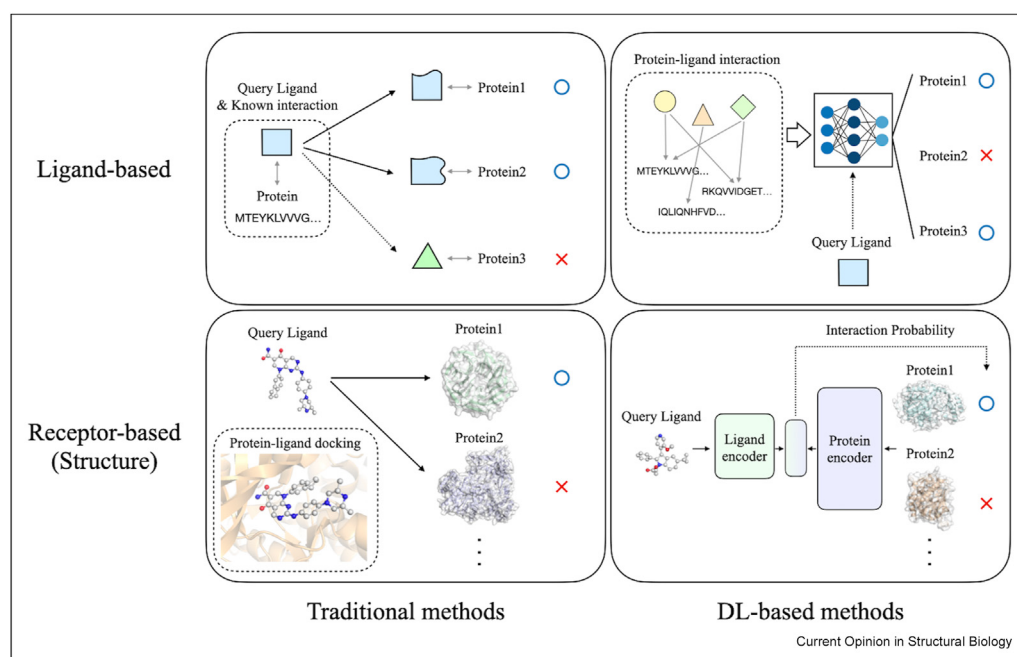
different flavors of molecular representations and DL architectures. For example, in the early stage, 1-D representations with a convolutional neural network (CNN) architecture [22] were taken, then later approaches were published using modern DL techniques [23]. DTI has the strength of using large amounts of training data, but also the limitation on the prediction resolution; for instance, on a recent benchmark on kinase screening, nonstructure-based DTI approaches generally performed worse than structure-based ones across metrics and test sets [24].

Receptor structure based approaches

Early-stage approaches using receptor structure information were based on molecular docking. Li *et al.* [61] was one of those early examples, and later GalaxySagittarius [2] was published. These approaches had the benefit over ligand-based approaches of providing structural bases of interactions, but remained rather minor because of the requirement of atomic-level receptor structures.

In the DL era, progress in structure-based approaches has been made from two directions. First is the increase in the size of receptor structure database. The applicability of the original approach has been expanded accordingly by including DL-predicted protein structures [62]. Second is the introduction of advanced DL techniques. Modern techniques from protein structure prediction, such as attention mechanisms or graph

Figure 2



Schematic view of receptor screening and method types.

representations, have been experimented for structure-based networks [25,26]. Despite their work, a majority of structure-based PLI networks are more focused on docking to date [63–65], which we expect will be developed for screening tools in the near future.

Receptor type-specific studies

A list of studies has focused on specific protein families instead of the entire proteome. Those works are still worth mentioning because of their efficiency for particular aims. Studies on certain families are relevant for drug repurposing or side-effect estimations. Kinases, important cancer therapy targets are most well studied in this endeavor. In several case studies, approved drugs were screened against hundreds of kinases to reveal novel kinase mechanisms [66]. Another group of families were studied for virtual toxicity screening to aid clinical trials. Examples are cytochrome P450 (CYP) and the hERG; their unexpected inhibition can cause significant adverse effects such as cardiotoxicity or hepatotoxicity. There are broad interests in developing methods specific for inverse screening against those receptor families [67–69].

Discussions

Many studies have claimed that their methods can now effectively screen the majority of both interacting and noninteracting pairs under their specific validation settings. However, there is substantial evidence suggesting that these claims are overestimates, resulting from particular validation conditions [70]. We emphasize two critical areas in dataset preparation that need improvement to develop more generalizable proteome-scale screening models.

First, data labeling needs improvement. Interactions are often transient and context-dependent. Experimental techniques are also known to be vulnerable to false positives. The authors of Topsy-Turvy suggest their performance gains over D-SCRIPT may come from GLIDE scores reducing the impact of mislabeled data. Establishing a clearer definition of ‘interaction’ would be helpful to reduce false labels. For example, STRING database [30] identifies functional associations as interactions, while many methods focus on physical interactions. Distinguishing these definitions could clarify the applications of different models. Similar issues occur in PLI datasets. Experimental values determined at different contexts (e.g. IC₅₀, EC₅₀, K_i, K_d) are collected and used together with less caution. Also, setting criteria for binders from

nonbinders is often arbitrary, as experimental values tend to vary continuously.

Second, the validation setting should be improved. Bennett et al. [70] argue that the current estimates of PPI model performances are inflated. They provide evidence that when the train/test split properly accounts for data leakage, most approaches fail to extract physically meaningful features and predict interactions randomly. Similarly, in PLI, it is shown that broadly used nonbinding data (called decoys) possesses artifacts that can be trivially captured [71]. In both fields, data leakage from wrong data split has been a common reason for the performance overestimation.

Although still in its early stages, structure-based deep learning approaches hold significant potential. For example, in the case of PPI prediction, it has been shown that conducting an initial screening using nonstructure-based methods, which allow for rapid, high-throughput analysis, followed by a structure-based screening, is an efficient approach [5]. A similar approach has been tested for PLI as well [72]. These approaches not only enable large-scale prediction but also provide additional insights based on complex structural information.

However, despite its promise, this approach remains challenging for certain types of PPIs or PLIs due to data limitations. To determine the final PPI structure accurately, information about stoichiometries is required. Several methods [73,74] have been developed to solve this problem, but their applicability and accuracy is still limited. Additionally, PPIs involving intrinsically disordered regions are underrepresented in the Protein Data Bank (PDB). Data limitation is more pronounced in the PLI study. There are only about 64,000 nonredundant PLI complex structures available even with a quite tight similarity filter using exact pair index match [39], and these are dominantly ‘target-centric,’ where various binders are bound to just a few protein families, such as kinases.

A critical challenge lies in how to secure sufficient structural datasets to support broad model training. Once highly accurate protein complex structure prediction for both PPIs and PLIs is achieved, the issue of target coverage in the structural dataset could be addressed through a self-distillation approach [3]. Recent advancements in deep learning-based complex structure prediction suggest that this vision may soon become a reality [65].

Declaration of competing interest

The authors declare no conflict of interest.

Acknowledgements

M.S.Y, K.K, and M.B. are supported by IITP/MSIT (RS-2023-00220628), NRF/MSIT (RS-2023-00210147, RS-2024-00440824). B.B and H.P are supported by NRF (2022R1C1C1007817).

Data availability

No data was used for the research described in the article.

References

Papers of particular interest, published within the period of review, have been highlighted as:

* of special interest

** of outstanding interest

- Singh R, Park D, Xu J, Hosur R, Berger B: **Struct2Net: a web service to predict protein-protein interactions using a structure-based approach**. *Nucleic Acids Res* 2010, **38**: W508–W515.
 - Yang J, Kwon S, Bae S-H, Park KM, Yoon C, Lee J-H, Seok C: **GalaxySagittarius: structure- and similarity-based prediction of protein targets for druglike compounds**. *J Chem Inf Model* 2020, **60**:3246–3254.
 - Jumper J, Evans R, Pritzel A, Green T, Figurnov M, Ronneberger O, Tunyasuvunakool K, Bates R, Zidek A, Potapenko A, *et al.*: **Highly accurate protein structure prediction with AlphaFold**. *Nature* 2021, **596**:583–589.
 - Baek M, DiMaio F, Anishchenko I, Dauparas J, Ovchinnikov S, Lee GR, Wang J, Cong Q, Kinch LN, Schaeffer RD, *et al.*: **Accurate prediction of protein structures and interactions using a three-track neural network**. *Science* 2021, **373**:871–876.
 - Humphreys IR, Pei J, Baek M, Krishnakumar A, Anishchenko I, Ovchinnikov S, Zhang J, Ness TJ, Banjade S, Bagde SR, *et al.*: **Computed structures of core eukaryotic protein complexes**. *Science* 2021, **374**, eabm4805.
 - Pei J, Zhang J, Cong Q: **Human mitochondrial protein complexes revealed by large-scale coevolution analysis and deep learning-based structure modeling**. *Bioinformatics* 2022, **38**: 4301–4311.
 - Humphreys IR, Zhang J, Baek M, Wang Y, Krishnakumar A, Pei J, Anishchenko I, Tower CA, Jackson BA, Warrior T, *et al.*: **Protein interactions in human pathogens revealed through deep learning**. *Nat Microbiol* 2024, **9**:2642–2652.
- This study describes a PPI prediction pipeline that utilizes structure prediction deep learning models. The authors analyzed 78 million protein pairs from 19 human pathogens, identifying 1923 complexes involving essential genes and 256 with virulence factors. Experimental validation confirmed half of the 12 tested predictions, offering insights into pathogen biology and potential drug targets.
- Sledzieski S, Singh R, Cowen L, Berger B: **D-SCRIPT translates genome to phenome with sequence-based, structure-aware, genome-scale predictions of protein-protein interactions**. *Cell Syst* 2021, **12**:969–982.e6.
 - Szyborski J, Emad A: **RAPPID: towards generalizable protein interaction prediction with AWD-LSTM twin networks**. *Bioinformatics* 2022, **38**:3958–3967.
 - Singh R, Devkota K, Sledzieski S, Berger B, Cowen L: **Topsy-Turvy: integrating a global view into sequence-based PPI prediction**. *Bioinformatics* 2022, **38**:i264–i272.
 - Sledzieski S, Devkota K, Singh R, Cowen L, Berger B: **TT3D: leveraging precomputed protein 3D sequence models to predict protein-protein interactions**. *Bioinformatics* 2023, **39**.
 - Topsy-Turvy is combined with the 3Di alphabet introduced by Foldseek by concatenating the protein language model embeddings with 3Di one-hot encodings. The embeddings between the two query proteins are combined as in D-SCRIPT and trained with the Topsy-Turvy loss function, resulting in better performance against Topsy-Turvy and D-SCRIPT. This suggests that incorporating structural information ameliorates PPI probability prediction.
 - Ko YS, Parkinson J, Liu C, Wang W: **TUnA: an uncertainty-aware transformer model for sequence-based protein-protein interaction prediction**. *Briefings Bioinf* 2024, **25**, bbae359.
- TUnA utilizes ESM2 embeddings along with a Spectral-normalized Neural Gaussian Process method to account for uncertainty in PPI prediction resulting in improved performance compared to Topsy-Turvy and D-SCRIPT. The ESM2 embeddings are processed with two separate encoders, where the first encoder processes the two query proteins separately, while the second encoder processes the concatenated embeddings of the two proteins. Additionally, spectral normalization is applied to the hidden layers, and in place of the traditional fully connected layer, a Gaussian process layer is used.
- Petrey D, Zhao H, Trudeau SJ, Murray D, Honig B: **PrePPI: a structure informed proteome-wide database of protein-protein interactions**. *J Mol Biol* 2023, **435**, 168052.
- Pre-PPI provides a way to utilize precomputed structures for PPI screening by extracting structural features of the predicted query protein structures. The predicted structures of the query proteins are retrieved from the AlphaFold Protein Structure Database, and related protein complexes within the PDB are found with structure alignment. The structure-based sequence alignment is used as a feature for computing the PPI probability.
- Zhao H, Petrey D, Murray D, Honig B: **ZEPPI: proteome-scale sequence-based evaluation of protein-protein interaction models**. *Proc Natl Acad Sci U S A* 2024, **121**, e2400260121.
 - Evans R, O'Neill M, Pritzel A, Antropova N, Senior A, Green T, Zidek A, Bates R, Blackwell S, Yim J, *et al.*: **Protein complex prediction with AlphaFold-Multimer**. *bioRxiv* 2022, <https://doi.org/10.1101/2021.10.04.463034>.
 - Bryant P, Pozzati G, Elofsson A: **Improved prediction of protein-protein interactions using AlphaFold2**. *Nat Commun* 2022, **13**:1265.
- AlphaFold2 is widely used to predict the structure of protein complexes, detecting positive protein-protein interactions. The authors developed a new method to predict the quality of complex structures named pDockQ. The new method can identify interacting proteins with 51% accuracy at a 1% false positive rate.
- Kim A-R, Hu Y, Comjean A, Rodiger J, Mohr SE, Perrimon N: **Enhanced protein-protein interaction discovery via AlphaFold-multimer**. *bioRxiv* 2024, <https://doi.org/10.1101/2024.02.19.580970>.
 - Mischley V, Maier J, Chen J, Karanickolas J: **PPIscreenML: structure-based screening for protein-protein interactions using AlphaFold**. *Elife* 2024, **13**.
 - Singh R, Sledzieski S, Bryson B, Cowen L, Berger B: **Contrastive learning in protein language space predicts interactions between drugs and protein targets**. *Proc Natl Acad Sci USA* 2023, **120**, e2220778120.
 - Gao B, Qiang B, Tan H, Ren M, Jia Y, Lu M, Liu J, Ma W, Lan Y: **DrugCLIP: Contrastive Protein-Molecule Representation Learning for Virtual Screening** 2023.
 - Goldman S, Das R, Yang KK, Coley CW: **Machine learning modeling of family wide enzyme-substrate specificity screens**. *PLoS Comput Biol* 2022, **18**, e1009853.
 - Zhao Q, Zhao H, Zheng K, Wang J: **HyperAttentionDTI: improving drug-protein interaction prediction by sequence-based deep learning with attention mechanism**. *Bioinformatics* 2021, **38**:655–662.
 - Bai P, Miljković F, John B, Lu H: **Interpretable bilinear attention network with domain adaptation improves drug-target prediction**. *Nat Mach Intell* 2023, **5**:126–136.
 - D'Souza S, Prema KV, Balaji S, Shah R: **Deep learning-based modeling of drug-target interaction prediction incorporating binding site information of proteins**. *Interdiscip Sci* 2023, **15**: 306–315.

This study shows an example of implicitly incorporating structural information of receptors into a PLI prediction network. Benchmarking on kinase screening, authors show their approach using binding site information gives superior results to previous sequence similarity-based approaches.

25. Wang P, Zheng S, Jiang Y, Li C, Liu J, Wen C, Patronov A, Qian D, Chen H, Yang Y: **Structure-aware multimodal deep learning for drug-protein interaction prediction**. *J Chem Inf Model* 2022, **62**:1308–1317.
- This study has two contributions: first, an unbiased dataset named GalaxyDB was established for PLI benchmarking to mitigate dataset issues in the PLI screening. Second, the authors built a structure-aware DL model for receptor screening using the receptor contact map as an input.
26. Yazdani-Jahromi M, Yousefi N, Tayebi A, Kolanthai E, Neal CJ, Seal S, Garibay OO: **AttentionSiteDTI: an interpretable graph-based model for drug-target interaction prediction using NLP sentence-level relation classification**. *Briefings Bioinf* 2022, **23**, bbac272.
- The authors introduced an interpretable graph attention mechanism and took receptor pocket information as input to build a PLI prediction network. This work is one of a few methods in which receptor structure information was implicitly considered.
27. Berman HM, Westbrook J, Feng Z, Gilliland G, Bhat TN, Weissig H, Shindyalov IN, Bourne PE: **The protein Data Bank**. *Nucleic Acids Res* 2000, **28**:235–242.
28. Consortium The UniProt, Bateman A, Martin M-J, Orchard S, Magrane M, Ahmad S, Alpi E, Bowler-Barnett EH, Britto R, Bye-A-Jee H, et al.: **UniProt: the universal protein knowledgebase in 2023**. *Nucleic Acids Res* 2022, **51**:D523–D531.
29. Oughtred R, Rust J, Chang C, Breitkreutz B-J, Stark C, Willems A, Boucher L, Leung G, Kolas N, Zhang F, et al.: **The BioGRID database: a comprehensive biomedical resource of curated protein, genetic, and chemical interactions**. *Protein Sci* 2021, **30**:187–200.
30. Szklarczyk D, Kirsch R, Koutrouli M, Nastou K, Mehryar F, Hachilif R, Gable AL, Fang T, Doncheva NT, Pyysalo S, et al.: **The STRING database in 2023: protein-protein association networks and functional enrichment analyses for any sequenced genome of interest**. *Nucleic Acids Res* 2023, **51**:D638–D646.
31. Luck K, Kim D-K, Lambourne L, Spirohn K, Begg BE, Bian W, Brignall R, Cafarelli T, Campos-Laborie FJ, Charleatoux B, et al.: **A reference map of the human binary protein interactome**. *Nature* 2020, **580**:402–408.
32. Zdravil B, Felix E, Hunter F, Manners EJ, Blackshaw J, Corbett S, de Veij M, Ioannidis H, Lopez DM, Mosquera JF, et al.: **The ChEMBL Database in 2023: a drug discovery platform spanning multiple bioactivity data types and time periods**. *Nucleic Acids Res* 2023, **52**:D1180–D1192.
33. Kim S, Chen J, Cheng T, Gindulyte A, He J, He S, Li Q, Shoemaker BA, Thiessen PA, Yu B, et al.: **PubChem 2023 update**. *Nucleic Acids Res* 2022, **51**:D1373–D1380.
34. Knox C, Wilson M, Klinger CM, Franklin M, Oler E, Wilson A, Pon A, Cox J, Chin NE, Lucy, Strawbridge SA, et al.: **DrugBank 6.0: the DrugBank knowledgebase for 2024**. *Nucleic Acids Res* 2023, **52**:D1265–D1275.
35. Liu Z, Su M, Han L, Liu J, Yang Q, Li Y, Wang R: **Forging the basis for developing protein–ligand interaction scoring functions**. 2017, <https://doi.org/10.1021/acs.accounts.6b00491>.
36. Wei H, Wang W, Peng Z, Yang J: **Q-BioLiP: a comprehensive resource for quaternary structure-based protein–ligand interactions**. *genom proteom bioinform* 2024, **22**, qzae001.
37. Davis MI, Hunt JP, Herrgard S, Ciceri P, Wodicka LM, Pallares G, Hocker M, Treiber DK, Zarinkar PP: **Comprehensive analysis of kinase inhibitor selectivity**. *Nat Biotechnol* 2011, **29**: 1046–1051.
38. Szklarczyk D, Santos A, von Mering C, Jensen LJ, Bork P, Kuhn M: **Stitch 5: augmenting protein-chemical interaction networks with tissue and affinity data**. *Nucleic Acids Res* 2016, **44**:D380–D384.

39. Durairaj J, Adeshina Y, Cao Z, Zhang X, Oleinikovas V, Duignan T, McClure Z, Robin X, Studer G, Kovtun D, et al.: **PLINDER: the protein-ligand interactions dataset and evaluation resource**. *bioRxiv* 2024, <https://doi.org/10.1101/2024.07.17.603955>.
40. Gilson MK, Liu T, Baitaluk M, Nicola G, Hwang L, Chong J: **BindingDB in 2015: a public database for medicinal chemistry, computational chemistry and systems pharmacology**. *Nucleic Acids Res* 2016, **44**:D1045–D1053.
41. Tran-Nguyen V-K, Jacquemard C, Rognan D: **LIT-PCBA: an unbiased data set for machine learning and virtual screening**. *J Chem Inf Model* 2020, <https://doi.org/10.1021/acs.jcim.0c00155>.
42. Mysinger MM, Carchia M, Irwin JJ, Shoichet BK: **Directory of useful decoys. Enhanced (DUD-E): Better Ligands and Decoys for Better Benchmarking** 2012, <https://doi.org/10.1021/jm300687e>.
43. Collins SR, Kemmeren P, Zhao X-C, Greenblatt JF, Spencer F, Holstege FCP, Weissman JS, Krogan NJ: **Toward a comprehensive atlas of the physical interactome of *Saccharomyces cerevisiae***. *Mol Cell Proteomics* 2007, **6**:439–450.
44. Berggård T, Linse S, James P: **Methods for the detection and analysis of protein–protein interactions**. *Proteomics* 2007, **7**: 2833–2842.
45. Ma N, Zhang Z-M, Lee J-S, Cheng K, Lin L, Zhang D-M, Hao P, Ding K, Ye W-C, Li Z: **Affinity-based protein profiling reveals cellular targets of photoreactive anticancer inhibitors**. *ACS Chem Biol* 2019, **14**:2546–2552.
46. Pai MY, Lomenick B, Hwang H, Schiestl R, McBride W, Loo JA, Huang J: **Drug affinity responsive target stability (DARTS) for small-molecule target identification**. *Methods Mol Biol* 2015, **1263**:287–298.
47. Yu Y, Anjum R, Kubota K, Rush J, Villen J, Gygi SP: **A site-specific, multiplexed kinase activity assay using stable-isotope dilution and high-resolution mass spectrometry**. *Proc Natl Acad Sci USA* 2009, **106**:11606–11611.
48. **Homogeneous time-resolved fluorescence assay to probe folded G protein-coupled receptors**. In *Methods in enzymology*. Academic Press; 2013:169–189.
49. Chen M, Ju C-J, Zhou G, Chen X, Zhang T, Chang K-W, Zaniolo C, Wang W: **Multifaceted protein-protein interaction prediction based on Siamese residual RCNN**. *Bioinformatics* 2019, **35**:i305–i314.
50. Bepler T, Berger B: *Learning protein sequence embeddings using information from structure*. 2019.
51. Lin Z, Akin H, Rao R, Hie B, Zhu Z, Lu W, Smetanin N, Verkuil R, Kabeli O, Shmueli Y, et al.: **Evolutionary-scale prediction of atomic-level protein structure with a language model**. *Science* 2023, **379**:1123–1130.
52. Devkota K, Murphy JM, Cowen LJ: **GLIDE: combining local methods and diffusion state embeddings to predict missing interactions in biological networks**. *Bioinformatics* 2020, **36**: i464–i473.
53. Hu J, Li Z, Rao B, Thafar MA, Arif M: **Improving protein-protein interaction prediction using protein language model and protein network features**. *Anal Biochem* 2024, **693**, 115550.
54. van Kempen M, Kim SS, Tumescheit C, Mirdita M, Lee J, Gilchrist CLM, Söding J, Steinegger M: **Fast and accurate protein structure search with Foldseek**. *Nat Biotechnol* 2024, **42**: 243–246.
55. Green AG, Elhabashy H, Brock KP, Maddamsetti R, Kohlbacher O, Marks DS: **Large-scale discovery of protein interactions at residue resolution using co-evolution calculated from genomic sequences**. *Nat Commun* 2021, **12**: 1–12.
56. Fenster JA, Azzinaro PA, Dinobol M, Borca MV, Spinard E, Gladue DP: **African swine fever virus protein–protein interaction prediction**. *Viruses* 2024, **16**:1170.

57. Zhang J, Pei J, Durham J, Bos T, Cong Q: **Computed cancer interactome explains the effects of somatic mutations in cancers.** *Protein Sci* 2022, **31**, e4479.
 58. Bennett NR, Coventry B, Goreshnik I, Huang B, Allen A, Vafeados D, Peng YP, Dauparas J, Baek M, Stewart L, *et al.*: **Improving de novo protein binder design with deep learning.** *Nat Commun* 2023, **14**:2625.
 59. Huang K, Xiao C, Glass LM, Sun J: **MolTrans: molecular Interaction Transformer for drug–target interaction prediction.** *Bioinformatics* 2020, **37**:830–836.
 60. Lee I, Keum J, Nam H: **DeepConv-DTI: prediction of drug-target interactions via deep learning with convolution on protein sequences.** *PLoS Comput Biol* 2019, **15**, e1007129.
 61. Li H, Gao Z, Kang L, Zhang H, Yang K, Yu K, Luo X, Zhu W, Chen K, Shen J, *et al.*: **TarFisDock: a web server for identifying drug targets with docking approach.** *Nucleic Acids Res* 2006, **34**:W219–W224.
 62. GalaxySagittarius-AF: **Predicting targets for drug-like compounds in the extended human 3D proteome.** *J Mol Biol* 2024, ******.
- This work highlights to what extent ligand-binding receptor screening can benefit from the advancement in the recent structure prediction networks. The method itself was inherited from the original GalaxySagittarius, but its applicability has been broadly expanded by taking AlphaFold models for the entire human proteome.
63. Corso G, Stärk H, Jing B, Barzilay R, Jaakkola T: **DiffDock. Diffusion Steps, Twists, and Turns for Molecular Docking** 2022.
 64. Krishna R, Wang J, Ahern W, Sturmfels P, Venkatesh P, Kalvet I, Lee GR, Morey-Burrows FS, Anishchenko I, Humphreys IR, *et al.*: **Generalized biomolecular modeling and design with RoseT-TAFold All-Atom.** *Science* 2024, **384**, ead12528.
 65. Abramson J, Adler J, Dunger J, Evans R, Green T, Pritzel A, Ronneberger O, Willmore L, Ballard AJ, Bambrick J, *et al.*: **Accurate structure prediction of biomolecular interactions with AlphaFold 3.** *Nature* 2024, **630**:493–500.
 66. Preynat-Seauve O, Nguyen EB-V, Westermaier Y, Héritier M, Tardy S, Cambet Y, Feyeux M, Caillon A, Scapozza L, Krause K-H: **Novel mechanism for an old drug: phenazopyridine is a kinase inhibitor affecting autophagy and cellular differentiation.** *Front Pharmacol* 2021, **12**, 664608.
 67. **Prediction of potential toxicity and side effect protein targets of a small molecule by a ligand–protein inverse docking approach.** *J Mol Graph Model* 2001, **20**: 199–218.
 68. Weiser B, Genzling J, Burai-Patrasu M, Rostaing O, Moitessier N: **Machine learning-augmented docking. 1. CYP inhibition prediction.** *Dig Dis* 2023, **2**:1841–1849.
 69. Creanza TM, Delre P, Ancona N, Lentini G, Saviano M, Mangiatordi GF: **Structure-based prediction of hERG-related cardiotoxicity: a benchmark study.** *J Chem Inf Model* 2021, <https://doi.org/10.1021/acs.jcim.1c00744>.
 70. Bernett J, Blumenthal DB, List M: **Cracking the black box of deep sequence-based protein–protein interaction prediction.** *Briefings Bioinf* 2024, **25**, bbae076.
 71. Chen L, Cruz A, Ramsey S, Dickson CJ, Duca JS, Hornak V, Koes DR, Kurtzman T: **Hidden bias in the DUD-E dataset leads to misleading performance of deep learning in structure-based virtual screening.** *PLoS One* 2019, **14**, e0220113.
 72. Gorgulla C, Nigam A, Koop M, Çönaroğlu SS, Secker C, Haddadnia M, Kumar A, Malets Y, Hasson A, Li M, *et al.*: **VirtualFlow 2.0 - the next generation drug discovery platform enabling adaptive screens of 69 billion molecules.** *bioRxiv* 2023, <https://doi.org/10.1101/2023.04.25.537981>.
 73. Kshirsagar M, Meller A, Humphreys I, Sledzieski S, Xu Y, Dodhia R, Horvitz E, Berger B, Bowman G, Ferres JL, *et al.*: **Rapid and accurate prediction of protein homo-oligomer symmetry with Seq2Symm.** *Res Sq* 2024, <https://doi.org/10.21203/rs.3.rs-4215086/v1>.
 74. Avraham O, Tsaban T, Ben-Aharon Z, Tsaban L, Schueler-Furman O: **Protein language models can capture protein quaternary state.** *BMC Bioinf* 2023, **24**:1–18.